

Effect of Preplant Fertilizer on Agronomic and Physiological Traits of Soybean Cultivars from Different Breeding Programs

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Abstract

Understanding the changes in agronomic and physiological traits associated with yield genetic gain is important for soybean production and future breeding strategy. The objective of this study was to compare the older and modern cultivars to learn whether the yield improvements depend on preplant fertilizer or the plant productivity improvement. A set of older cultivars, with their modern counterparts derived from breeding programs in Liaoning and Ohio were evaluated for their agronomic and physiological traits under different fertilizer levels from 2004 to 2006. There was no improvement of response to N and P preplant fertilizer for genotypes. After more than 70 yr breeding, soybean breeders made some improvements in agronomic and physiological traits that contribute to yield increase. When compared to older cultivar, modern Liaoning and Ohio cultivars were shorter and more resistance to lodging, had greater leaf density, higher harvest index, more leaf area per plant, and greater photosynthetic rate, transpiration rate and stomatal conductance at the beginning of seed development. Ohio cultivars were more resistant to lodging as if selected for easy harvest by combine, even under high N and P preplant fertilizer level, which resulted in Ohio cultivars with higher and stable yield productivity.

Key words: soybean [*Glycine max* (L.) Merr.], fertilizer, agronomic trait, physiology, breeding

INTRODUCTION

Understanding the changes in agronomic and physiological traits associated with yield genetic gains is important for the production and future breeding strategy of soybean [*Glycine max* (L.) Merr.]. Soybean yield genetic improvement in North America has been reported to be in the range of 0.5 to 1% annually (Wilcox *et al.* 1979; Voldeng *et al.* 1997). The improvement of several physiological traits during six decades (1934-1992) of cultivar releases in Ontario resulted in a yield genetic gain of about 0.5% per year. Some traits such

as lodging improved, but in others (greater N₂-fixation, greater stress tolerance) more subtly (Specht *et al.* 1999). Cober *et al.* (2005) reported that new soybean cultivars had a stronger tolerance to plant population stress than older cultivars. There was a interaction of release year $\times$ year, the newer cultivars yielded more under the highest yielding years (Specht *et al.* 1999). Cober *et al.* (2005) also found that seed yield differences existed among experimental years. Liu *et al.* (2005) reported that the seed number per pod increased in high rainfall year but had no relationship with the variation in yield of the genotypes evaluated.

Some researchers concluded that soybean breeders

Received 21 April, 2010 Accepted 19 July, 2010

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have selected for higher photosynthetic rate as they selected for higher yield (Dornhoff and Shibbles 1970; Buttery and Buzzell 1972; Ojima 1972; Wells *et al.* 1982). There was a significant correlation between leaf photosynthetic rates during seed filling and seed yield (Buttery *et al.* 1981). Boerma and Ashley (1988) also indicated that high photosynthetic capacity and long seed-fill period would result in high seed yield. In Canada soybean yield gain over 58 yr (1934–1992) was associated with the improvement of harvest index, photosynthetic rate and stomatal conductivity; compared with the older cultivars, modern cultivars were more efficient at producing and allocating carbon resources (Morrison *et al.* 1999).

Chlorophyll concentration or leaf greenness was correlated with photosynthetic rate (Buttery *et al.* 1981; Ma *et al.* 1995), but not with yield (Buttery *et al.* 1981). However, Morrison *et al.* (1999) reported that increased chlorophyll concentration was not correlated with higher leaf photosynthesis. No significant relationship existed between leaf chlorophyll concentration or leaf greenness and year of cultivar release (Ma *et al.* 1995; Morrison *et al.* 1999).

There was a significant contribution of harvest index to yield improvement (Frederick *et al.* 1991; Morrison *et al.* 1999), and after the beginning of the seed-filling period the difference of dry matter accumulation between old and modern Japanese genotypes was most apparent (Shiraiwa and Hashikawa 1995). Kumudini *et al.* (2001) concluded that seed yield of the new cultivars was 30% greater than their older counterparts, the increase in dry matter accumulation contributed 78% and of harvest contributed 22% towards the yield genetic gain. Liu *et al.* (2005) found that there was a significant relationship among increased dry matter accumulation, higher leaf area index, and leaf area duration during reproductive stages and high-yielding genotypes in each maturity group tested. No relationship existed between harvest index and seed yield. Kumudini (2002) proposed that there was a cultivar-specific source-sink ratio, which may be a more potent trait than assimilate supply alone for yield improvement. Specht *et al.* (1999) indicated that recent cultivars seemed to be superior to older ones in the supply of photosynthate to sinks across a wide range of environments.

Recently, researchers have been interested in the application of fertilizers for soybean production. Wesley *et al.* (1999) reported that application of N at the beginning pod growth stage in Kansas increased yield at four irrigated sites, but had no effect on the content of seed protein or oil. Afza *et al.* (1987) applied N during the R4 (Fehr and Caviness 1977) growth stage of soybean, and got a seed yield increase. Other researchers concluded that N applied during the reproductive stages did not result in a yield increase and was not a recommended practice (Schmitt *et al.* 2001; Barker and Sawyer 2005). Sij *et al.* (1979) postulated that early vegetative growth could be stimulated by a small amount of N applied at planting, however, the results of most studies that used N fertilizer as ‘Starter N’ showed that soybean grown on most soils had no response to low rates (25 to 35 kg N ha⁻¹) of preplant N fertilization (Sij *et al.* 1979; Johnson 1987; Heatherly *et al.* 2003). Kumudini *et al.* (2002) reported that genetic improvement of the short-season soybeans was a consequence of continued N accumulation during the seed filling period and was not related to differences of remobilizing or partitioning N to the seed among the genotypes.

Soybean as a monoculture crop or planted where the preceding wheat *Triticum aestivum* L. did not receive any fertilizer P had a significant response to application of P up to 80 kg P₂O₅ ha⁻¹ (Aulakh *et al.* 1990, 2003). Bullen *et al.* (1983) reported that applying P in a band near the seed was superior to broadcasting application for soybean yield increase. Compared with deep placement, broadcast application of P improved soybean yield (Lutz and Jones 1974). Ham and Caldwell (1978) reported that fertilizer P increased soybean production but placement had no significant effect on yield and P uptake.

Improved agronomic practices or soybean genetic gain or the interaction of the two may result in greater seed yield (Evans 1993). It is not known whether breeders have inadvertently altered the agronomic and physiological responsiveness of cultivars to N and P preplant fertilizer. The objective of this study was to measure agronomic and physiological traits of older and modern soybean cultivars supplied with different rates of N and P fertilizer to determine the source of seed yield improvements.

MATERIALS AND METHODS

We compared 12 cultivars with similar maturity (between MG III and MG IV), including four old cultivars that were introduced into the USA from Liaoning Province of China before extensive North American breeding efforts began. The four older cultivars (Shingto, Mukden, Harbinsoy, and Boone) were named and released in the USA, but of these only Mukden was ever widely grown. Mukden is also the only cultivar of the four that contributed significantly as an ancestor of current USA cultivars. The remaining eight cultivars in our tests were developed recently at Liaoning Province, China (Liaodou 11, Liaodou 12, Shennong 94-11, and Shendou 4) and The Ohio State University, USA (HS93-

4118, Ohio FG1, Darby, and Kottman). The basis for the choice of older cultivars for this experiment was geographical rather than pedigree: the cultivars were collected in Liaoning and presumably grown there. Except for Mukden they were not ancestral to the Ohio cultivars, and there was no evidence that they were ancestral to modern Liaoning cultivars. We are assuming that they typify soybean germplasm grown in Liaoning, 85 to 100 yr ago.

We conducted the experiment in the field under rainfed conditions and without insect and disease stress at latitude approximately 40°N at Shenyang, Liaoning for 3 yr from 2004 to 2006. The soil characteristics to 20 cm depth are shown in Table 1.

Available N was extracted by alkaline KMnO_4 , avail-

Table 1 Soil characteristics of experimental locations¹⁾

Year	Former crop	Soil type	Organic matter (g kg ⁻¹)	Total N (mg kg ⁻¹)	Available N (mg kg ⁻¹)	Available P (mg P ₂ O ₅ kg ⁻¹)	Available K (mg K ₂ O kg ⁻¹)	pH
2004	Maize	Loam	16.7	852.1	85.8	21.2	121.5	7.4
2005	Maize	Loam	13.8	790.0	75.4	16.3	75.5	7.3
2006	Maize	Loam	19.5	967.0	96.6	13.6	116.4	7.5

¹⁾ Soil sample depth was 20 cm.

able P by the Olsen method (Olsen *et al.* 1954), and available K by NH_4OAc . Diammonium hydrogen phosphate $(\text{NH}_4)_2\text{HPO}_4$ was applied at three treatments: no fertilizer treatment as a check, 150 kg ha⁻¹ as the middle level treatment, 300 kg ha⁻¹ as the high level treatment. A split-plot design arranged in randomized complete blocks was used with three replications. Cultivars were the main-plot factor, and the split-plot factor consisted of three levels of fertilizer. The N-P fertilizer (N18%-P₂O₅46%) was used as a preplant fertilizer, applied in a band near the seeds at the same depth. The plots consisted of five rows, 5 m in length spacing 60 cm apart. The distance between plants within a row was 11 cm and plant density (commonly used by farmers in Liaoning) was 150 000 plants ha⁻¹. The plant density was achieved by over-planting and thinning. Seeds without inoculation were planted in April 28, 2004 and 2005, and May 1, 2006. 2 d after planting, herbicides were applied for weed control.

During the growing season, at R2, R4 and R5 growth stages (Fehr and Caviness 1977) apparent photosynthetic rate, transpiration rate and stomatal conductance of leaves were measured using a portable photosynthe-

sis system (Li-Cor Model 6400) with 6400-02B LED source in the morning (9-12 a.m.) of sunny day. The leaf chamber was 2 cm × 3 cm = 6 cm². Before each measurement, the instrument was self calibrated. During measurement the flow rate was 500 μmol s⁻¹, reference CO₂ 400 μmol mol⁻¹, block temperature 30°C, leaf temperature 30°C, light intensity 1 000 μmol m⁻² s⁻¹, and 15 seconds for the leaf to adjust to chamber conditions before data were collected. The fourth leaf from the top of the plant was used, and three leaves per plot were sampled.

Leaf greenness of the fourth leaf from the top of the plant at R1, R2 and R3 growth stages was measured using a chlorophyll meter (Model Spad502), and three leaves per plot were sampled.

Every year four plants were sampled for measuring leaf area and leaf density in the plots of first replication at R2, R4 and R5 growth stages. The leaf area was measured with a leaf area meter (Model-3000, Li-Cor), the leaf area per plant was divided by plant height at the same growth stage to determine the leaf density.

Before harvesting, ten consecutive plants in a row were selected from each plot for the measurement of

agronomic and morphological traits. Each sampled plant was measured for plant height from the node of cotyledon to the tip of stem, the nodes of the main stem and the number of branches were counted, and the seed yield of branches and main stem were measured separately. Seed yield of the main stem was divided by the combined seed yield of the main stem and branches to determine the percentage of yield produced on the main stem. Seed yield was divided by the total plant weight to determine the harvest index. After soybeans matured, the 3-m long middle part of the inside three rows was harvested for yield.

In the SAS statistical analysis of agronomic traits and photosynthesis rate, transpiration rate and stomatal conductance, years were considered as non-random factors. As the leaf area, leaf density were measured in only one replicate from each genotype $\times$ fertilizer rate $\times$ year, in the statistical analysis used three years as replicates to test the effects of genotype and fertilizer.

RESULTS

Agronomic traits

There was no significant effect of N and P preplant fertilizer on the yield (Table 2), and no fertilizer $\times$ cultivar interaction. There was a significant difference of seed yield among cultivars. Modern Ohio cultivars had the highest yield, followed by modern Liaoning cultivars, and the old Liaoning cultivars had the lowest yield.

The N and P preplant fertilizer had a significant effect on the percentage of yield produced on the main stem (Table 2); changes in the proportion of yield located in the main stem might be a consequence of the promotion of branch growth because of a higher initial

level of nutrients. As breeding programs have changed the proportion of seed weight on main stem and branches, there was a significant difference among the cultivars in percentage of seed yield on the main stem. Averaged across fertilizer levels of the modern Liaoning cultivars had the highest percentage (82%), followed by the modern Ohio cultivars (76%), and the old Liaoning cultivars (68%) (Table 2). This might be related with the growth habit of cultivars, modern Liaoning cultivars being semideterminate genotypes, and modern Ohio cultivars and old Liaoning cultivars indeterminate.

The N and P preplant fertilizer had no significant effect on the harvest index (Table 2), however after about 70 yr breeding improvement, modern Liaoning and Ohio cultivars had a higher harvest index compared with the old Liaoning cultivars.

The N and P preplant fertilizer significantly increased the plant height and there was no significant interaction between fertilizer and cultivar (Table 3). Fertilizer significantly increased lodging (Table 3), but the effect was more pronounced in some cultivars, giving a highly significant interaction between fertilizer and cultivar. Some cultivars (such as Ohio cultivars Darby and Kottman) showed more tolerance to lodging than the other cultivars.

Internode length could be an important morphological trait related to lodging. There was a significant difference among cultivars; old Liaoning cultivars had a greater internode length than the modern cultivars (Table 3). Fertilizer, however, had no significant influence on internode length, which implied that the increased lodging was caused by the effect of fertilizer on some other aspect of plant growth such as larger leaf area. Larger leaf area at R4 stage caused by initial fertilizer made the lodging more severe (Fig. 1).

There was a significant interaction of cultivar $\times$ year

Table 2 Effect of fertilizer on yield and its related traits of soybean cultivars released from different breeding programs

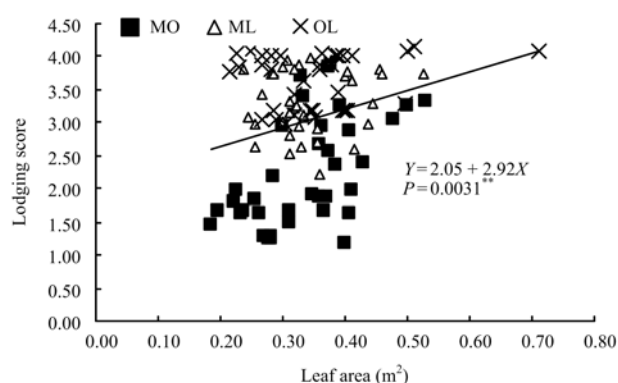
Group	Fertilizer level (kg ha ⁻¹)								
	0	150	300	0	150	300	0	150	300
	Yield (mg ha ⁻¹)			Percentage of yield on main stem (%)			Harvest index		
Modern Ohio	2.64 $\pm$ 0.09	2.64 $\pm$ 0.04	2.60 $\pm$ 0.10	76 $\pm$ 7.9	79 $\pm$ 9.0	74 $\pm$ 9.8	0.52 $\pm$ 0.02	0.50 $\pm$ 0.02	0.50 $\pm$ 0.01
Modern Liaoning	2.50 $\pm$ 0.13	2.58 $\pm$ 0.17	2.42 $\pm$ 0.21	85 $\pm$ 3.9	83 $\pm$ 4.0	80 $\pm$ 3.1	0.49 $\pm$ 0.02	0.47 $\pm$ 0.01	0.48 $\pm$ 0.03
Old Liaoning	1.73 $\pm$ 0.19	1.79 $\pm$ 0.13	1.71 $\pm$ 0.20	70 $\pm$ 16.7	69 $\pm$ 16.5	65 $\pm$ 19.3	0.42 $\pm$ 0.06	0.41 $\pm$ 0.08	0.41 $\pm$ 0.06
Mean of all cultivars	2.29 $\pm$ 0.44	2.34 $\pm$ 0.42	2.25 $\pm$ 0.44	77 $\pm$ 11.7	77 $\pm$ 11.7	73 $\pm$ 13.0	0.48 $\pm$ 0.05	0.46 $\pm$ 0.06	0.46 $\pm$ 0.06
P-value									
Cultivar	<0.0001**			<0.0001**			<0.0001**		
Fertilizer	0.0688			0.0217*			0.3347		
Cultivar $\times$ Fertilizer	0.9966			0.6154			0.5460		
Cultivar $\times$ Year	<0.0001**			<0.0001**			<0.0001**		

*, significant at 5% level; **, significant at 1% level.

Table 3 Effect of fertilizer on morphological traits of soybean cultivars released from different breeding programs

Group	Fertilizer level (kg ha ⁻¹)								
	0	150	300	0	150	300	0	150	300
	Plant height (cm)			Lodging ¹⁾ (score)			Internode length (cm)		
Modern Ohio	99±2.9	100±2.5	102±1.8	1.9±0.7	2.1±0.7	2.6±0.8	4.4±0.2	4.4±0.3	4.4±0.3
Modern Liaoning	118±12.9	123±14.2	124±13.3	2.9±0.3	3.5±0.1	3.5±0.1	4.8±0.1	4.9±0.1	4.9±0.1
Old Liaoning	147±39.6	153±45.2	153±41.8	3.6±0.2	3.7±0.2	3.7±0.1	6.0±1.0	6.2±1.2	6.1±1.0
Mean of all cultivars	122±30.1	125±33.5	126±31.7	2.8±0.8	3.1±0.8	3.3±0.7	5.0±0.9	5.1±1.0	5.1±0.9
<i>P</i> -value									
Cultivar	<0.0001**			<0.0001**			<0.0001**		
Fertilizer	0.0182*			0.0004**			0.2638		
Cultivar×Fertilizer	0.0808			<0.0001**			0.5747		
Cultivar×Year	<0.0001**			0.0003**			<0.0001**		

¹⁾ 1=erect, 5=prostrate.

**Fig. 1** Regression of leaf area per plant at R4 growth stage and lodging.

in all measured agronomic traits, which implied that soybean growth was very sensitive to the environments.

Physiological traits at different stages

At R2, R4, and R5 growth stages, there was no significant effect of N and P preplant fertilizer on the leaf apparent photosynthetic rate, transpiration rate, and stomatal conductance of soybean cultivars released from different breeding programs (Table 4). No significant interactions existed between fertilizer and cultivar in these physiological traits at R4 growth stage, but there were significant interactions in apparent photosynthetic

rate, transpiration rate, and stomatal conductance at R2 and R5 (no interaction for stomatal conductance at R5 stage). There were significant differences in the leaf apparent photosynthetic rate, transpiration rate and stomatal conductance among cultivars. The averages of photosynthetic rate, transpiration rate and stomatal conductance of modern Ohio and Liaoning cultivars were higher than those of the old Liaoning cultivars. There were significant linear regressions between the apparent photosynthetic rates at R2, R4 and R5 growth stages and seed yield. The correlations of apparent photosynthetic rate at R4 and R5 stages and seed yield were significant and positive, but at R2 stage the correlation was significant and negative. Higher apparent photosynthetic rate at the late reproductive stages resulted in higher seed yield (Fig. 2).

There was a significant linear regression between apparent photosynthetic rate and transpiration rate, stomatal conductance. The improvement in transpiration rate and stomatal conductance increased apparent photosynthetic rate.

There was no significant effect of N and P preplant fertilizer on leaf greenness at R1, R2 and R3 growth stages ($P_1=0.4739$, $P_2=0.0724$, $P_3=0.2382$), but there was a significant difference on leaf greenness among cultivars at R1, R2 and R3 growth stages ($P_1<0.0001^{**}$, $P_2<0.0001^{**}$, $P_3<0.0001^{**}$). The greenness of leaf had

Table 4 Effect of fertilizer on leaf apparent photosynthesis rate, transpiration rate, and stomatal conductance of soybean cultivars at different growth stages (*P*-value)

Growth stage	R2			R4			R5		
	Photosynthesis rate	Transpiration rate	Stomatal conductance	Photosynthesis rate	Transpiration rate	Stomatal conductance	Photosynthesis rate	Transpiration rate	Stomatal conductance
Cultivar	<0.0001**	<0.0001**	<0.0001**	<0.0001**	<0.0001**	<0.0001**	<0.0001**	<0.0001**	<0.0001**
Fertilizer	0.3075	0.2846	0.2910	0.0803	0.3355	0.4159	0.1638	0.2635	0.1314
Cultivar×Fertilizer	0.0001**	<0.0001**	0.0009**	0.3803	0.5218	0.2836	0.0114*	0.0001**	0.0547
Cultivar×Year	<0.0001**	<0.0001**	<0.0001**	<0.0001**	<0.0001**	<0.0001**	<0.0001**	<0.0001**	<0.0001**

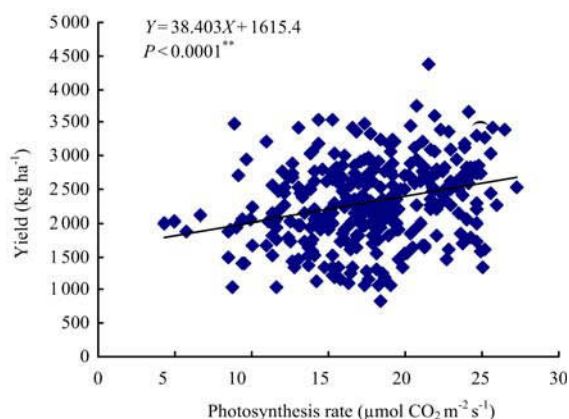


Fig. 2 Regression of photosynthesis rate at R5 growth stage and seed yield.

been improved by soybean breeding. For example, at R2 stage the average leaf greenness of modern Ohio and Liaoning cultivars was increased by 10.8 and 8.9%, respectively, compared with the old Liaoning cultivars. A significant interaction existed between fertilizer and cultivar at R1 and R3 growth stages ($P_1=0.0044^{**}$, $P_3<0.0001^{**}$).

There were significant interactions of cultivar $\times$ year for all measured physiological traits at R1, R2 and R5 growth stages. Some cultivars were more sensitive to environment than the others.

Leaf area and leaf density

There was no significant effect of N and P preplant fertilizer on the leaf area per plant at R2 and R5 growth stages, but there was a significant effect at R4 growth stage (Table 5). There was no interaction of cultivar $\times$ fertilizer at R2, R4 and R5 growth stages. There were significant differences in leaf area per plant among cultivars at R4 and R5 growth stages. Significant linear regressions existed between leaf area per plant and lodging score at R2 and R4 (Fig. 1) growth stages. The correlations of leaf area per plant and lodging score were significant, 0.30 ** and 0.28 * for R2 and R4 stages, respectively. Ohio cultivars had a smaller average leaf area (0.27 m²/plant) than modern and old Liaoning cultivars (both 0.29 m²/plant). Ohio selection with greater emphasis on lodging resistance gave more improve-

Table 5 Effect of fertilizer on leaf area per plant and leaf density of soybean cultivars released from different breeding programs at different growth stages (P -value)

Growth stage	R2		R4		R5	
	Leaf area	Leaf density	Leaf area	Leaf density	Leaf area	Leaf density
Cultivar	0.1354	0.0223 *	0.0224 *	0.0470 *	<0.0001 **	0.0002 **
Fertilizer	0.1049	0.0417 *	0.3052	0.2455	0.1921	0.1660
Cultivar $\times$ Fertilizer	0.5842	0.0740	0.7130	0.0272 *	0.1016	0.3368

ment in leaf area per plant than in the Liaoning breeding program. However at R5 growth stage, modern Ohio (0.27 m²/plant) and Liaoning (0.28 m²/plant) kept higher leaf area per plant than the old Liaoning cultivars (0.26 m²/plant). As mentioned previousing, the modern Ohio and Liaoning cultivars had a higher photosynthesis rate at R5 growth stage, which might be the one of causes resulting in modern cultivars with higher yield productivity. A significant quadratic regression existed between leaf area per plant at R5 growth stage and seed yield (Fig. 3), for the maximum seed yield the leaf area per plant should be 0.33 m² at R5 stage.

There was no significant effect of N and P preplant fertilizer on the leaf density at R2, R4 and R5 growth stages. There was a significant interaction of cultivar $\times$ fertilizer at R2, but no interaction at R4 and R5. A

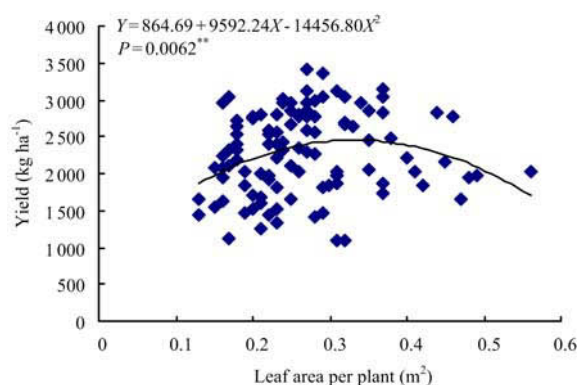


Fig. 3 Regression of leaf area per plant at R5 growth stage and seed yield.

significant difference existed in the leaf density among cultivars at R2, R4 and R5 growth stages (Table 5).

There were significant linear regressions between leaf density and seed yield at R2, R4, and R5 (Fig. 4) growth stages. The correlations of leaf density and seed yield were significant, 0.27**, 0.28**, and 0.45** for R2, R4, and R5 growth stages, respectively. Although Ohio cultivars had a smaller average leaf area (0.27 m²/plant) than modern and old Liaoning cultivars (both 0.29 m²/plant), they had the highest averaged leaf density (26.5 cm² cm⁻¹) compared with modern (21.6 cm² cm⁻¹) and old Liaoning cultivars (16.6 cm² cm⁻¹). Modern cultivars had a higher leaf density than old cultivars, which revealed that both breeding programs balanced the improvement of plant height and leaf area. Ohio cultivars had a smaller leaf area per plant and shorter plant height, which resulted in a higher leaf density. Modern Liaoning cultivars had a similar leaf area per plant to the old cultivars, but they had a greater leaf density as they had a shorter plant height.

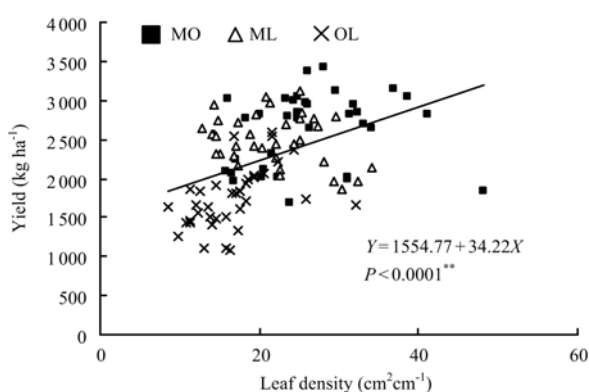


Fig. 4 Regression of leaf density at R5 growth stage and seed yield.

DISCUSSION

Soybean is particularly rich in N and P elements. However, under rainfed conditions, the yield potential of the cultivars may have not been expressed, the demand for nutrients was relatively low, and there was no improvement in response to N and P preplant fertilizer for genotypes. Tian *et al.* (2008) reported that decreasing N fertilizer rate at seeding time could reduce the inhibitory effect of nitrogen on root nodules, promote nutrient and dry matter accumulation, promote pod bearing and filling, and increase the yield. Com-

pared with basal application, topdressing of N at early pod filling stage increased yield by 3.33%, and there was no significant effect of topdressing N on soybean yield in blooming or at pod filling stages. Seed yield was increased by starter N in 2 out of the 3 yr, compared to the no N treatment, there was a seed increase of greater than 5% for the 16 kg N ha⁻¹ rate (Osborne and Riedell 2006). Gan *et al.* (1998, 2003) reported that starter N at 25 kg ha⁻¹ resulted in minimum yield in all three tested genotypes, and they concluded that flowering stage was the best timing for N top-dressing during reproduction. The optimal time of topdressing nitrogen for multiple soybean after wheat was at the late stage of podding (Li *et al.* 1995). A positive effect of starter fertilizer application on early plant biomass and plant N existed, but this increase of plant growth lasted at R3 stage (Osborne and Riedell 2006). Nitrogen as starter fertilizer could increase leaf area index and the dry matter accumulation, but there was no significant yield increment (Cai *et al.* 2007). In terms of yield formation, reproductive stages were more critical than vegetative ones. Modern cultivars yielded better because their photosynthesis was high during the critical period, and starter fertilization did not modify this response because its effects were mostly restricted to the vegetative period.

In the soil with low available P (less than 10 mg P₂O₅ kg⁻¹), application of P fertilizer increased soybean yield, however, there was no effect on yield if available P was over 20 mg P₂O₅ kg⁻¹ (Wang *et al.* 2006). The effect of phosphate fertilizer on soybean yield was related with the soil types, the yield was raised by 6.73, 9.56, and 7.69% in black, meadow and lessive soil, respectively (Li 2002). Under good P soil, topdressing of 18.75 kg N ha⁻¹ resulted in 6% yield increase, topdressing of N, P, and K fertilizers together, the yield increased significantly (Li *et al.* 2008).

Grain yield, leaf area, leaf area index, stem diameter, and specific leaf weight increased with the years of release (Zheng *et al.* 2006). Plant height and lodging score decreased with the increase in stem diameter, number of branches decreased with the increase in number of node, length of internode was increasingly shortened with resistance improvement of stem (Zhao *et al.* 2008). The seed yield of modern Ohio and Liaoning cultivars was significantly higher than that of old

Liaoning cultivars. Seed yield improvement was from the improvements of agronomic traits such as tolerant to lodging, high harvest index, and the photosynthetic activity at reproductive growth stages. There was a significant effect of N and P preplant fertilizer on the leaf area per plant at R4 growth stage, however, a significant linear regression existed between leaf area per plant at R4 stage and lodging score. The effect of fertilizer on leaf area at R4 stage did not result in the increase of seed yield as more lodging occurred.

Leaf density is an important canopy trait. High leaf density gives a poor shaded canopy, and low density means the lack of leaf area for photosynthesis. Keeping a relatively high leaf density at late reproductive stage is critical for yield. There was a significant difference in leaf density among cultivars at R2, R4, and R5 growth stages, and significant linear regressions existed between leaf density at R2, R4, and R5 stages and seed yield. Ohio cultivars had the highest average leaf density, although they had a smaller average leaf area per plant than modern and old Liaoning cultivars. Modern Ohio and Liaoning cultivars had a higher leaf density than old Liaoning cultivars, which revealed that both breeding programs balanced the improvement of plant height and leaf area and made the canopy more productive. Ohio cultivars had a smaller leaf area per plant and shorter plant height, which resulted in a higher leaf density. Modern Liaoning cultivars had a similar leaf area per plant to the old cultivars, but they had a greater leaf density as they had a shorter plant height.

Even though some old cultivars had high photosynthesis and stomatal conductance, they had high plant height and lodging, which resulted in low yield. For high seed yield, the good combination of agronomic trait and physiological trait is important.

CONCLUSION

Understanding the changes in traits associated with yield genetic gain is important for soybean production and future breeding strategy. The study indicated that in these environments, the yield potential of the cultivars may have not been expressed. The demand for nutrients was relatively low, with no improvement of response to N and P preplant fertilizer for genotypes. After more than 70 yr breeding, soybean breeders made

some improvements in agronomic and physiological traits that contribute to yield increase. The goals of soybean breeding programs in Ohio and Liaoning may be different in some traits but not in others, both of them are looking for higher yields. Modern Liaoning and Ohio cultivars had shorter plant height, more resistance to lodging, higher leaf density, higher harvest index, more leaf area per plant, and higher photosynthetic rate, transpiration rate, and stomatal conductance at R5 growth stage compared the old cultivars. Because of differences in the goals of soybean breeding programs in Liaoning, China and Ohio, USA, the released cultivars had some distinctive traits. Ohio cultivars were more resistant to lodging as if selected for easy harvest by combine, even under high N and P preplant fertilizer level, which resulted in Ohio cultivars with higher and stable yield productivity. A significant linear regression existed between leaf area per plant at R4 stage and lodging score, and significant linear regressions existed between leaf density at R2, R4, and R5 stages and seed yield, which revealed that the improvement of plant height and leaf area increased the productivity of soybean cultivars. For high seed yield, the good combination of agronomic traits and physiological traits is important.

Acknowledgements

The financial support for this research was provided by the state and federal funds of Ohio Agricultural Research and Development Center, The Ohio State University, USA.

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(Managing editor WANG Ning)